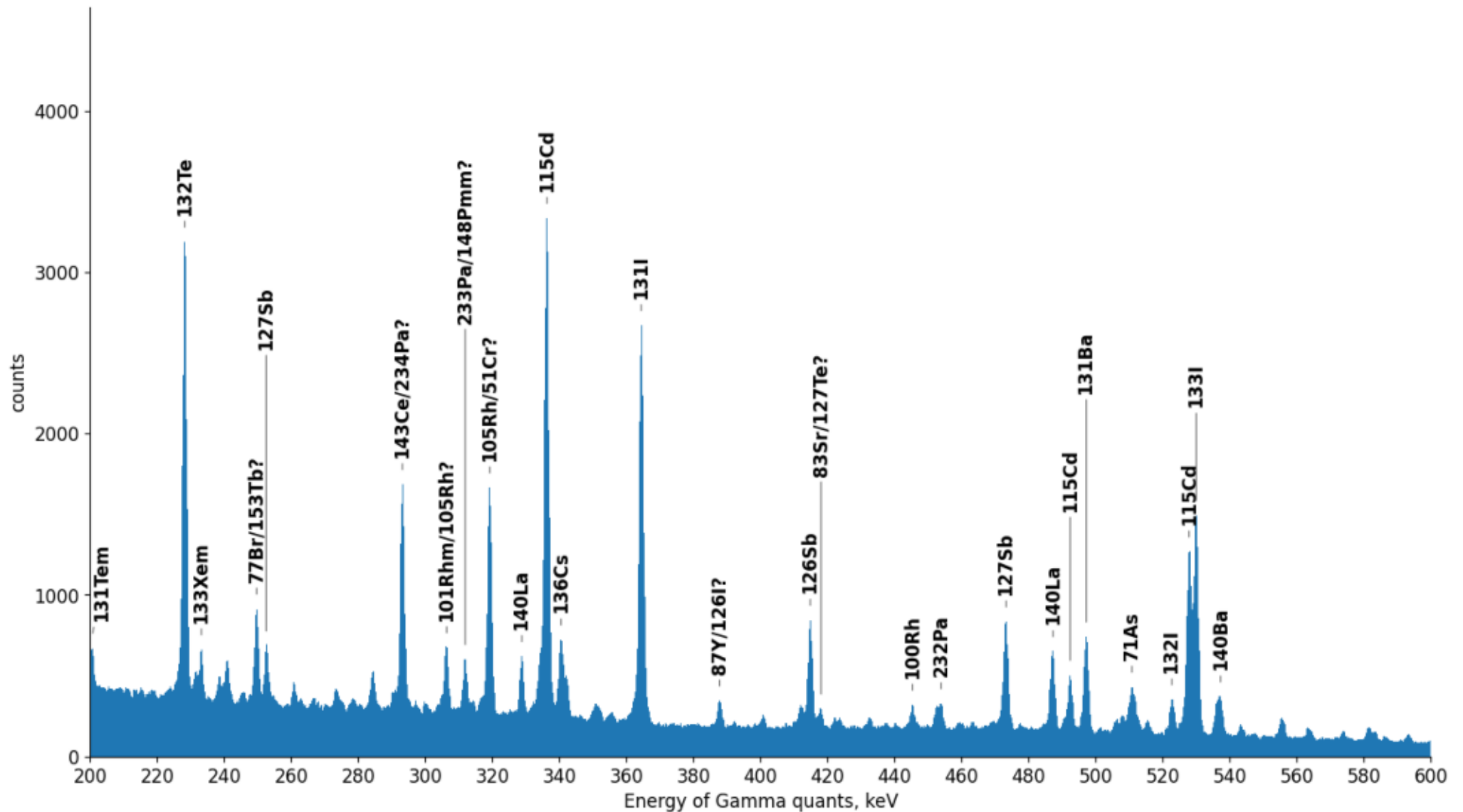


API-based tools — sample outputs

The gamma-spectrum analyzer and the nuclide chart viewer running on the public IAEA Live Chart of Nuclides API (nds.iaea.org) instead of the curated gamma library and the IsotopeDB MySQL database.

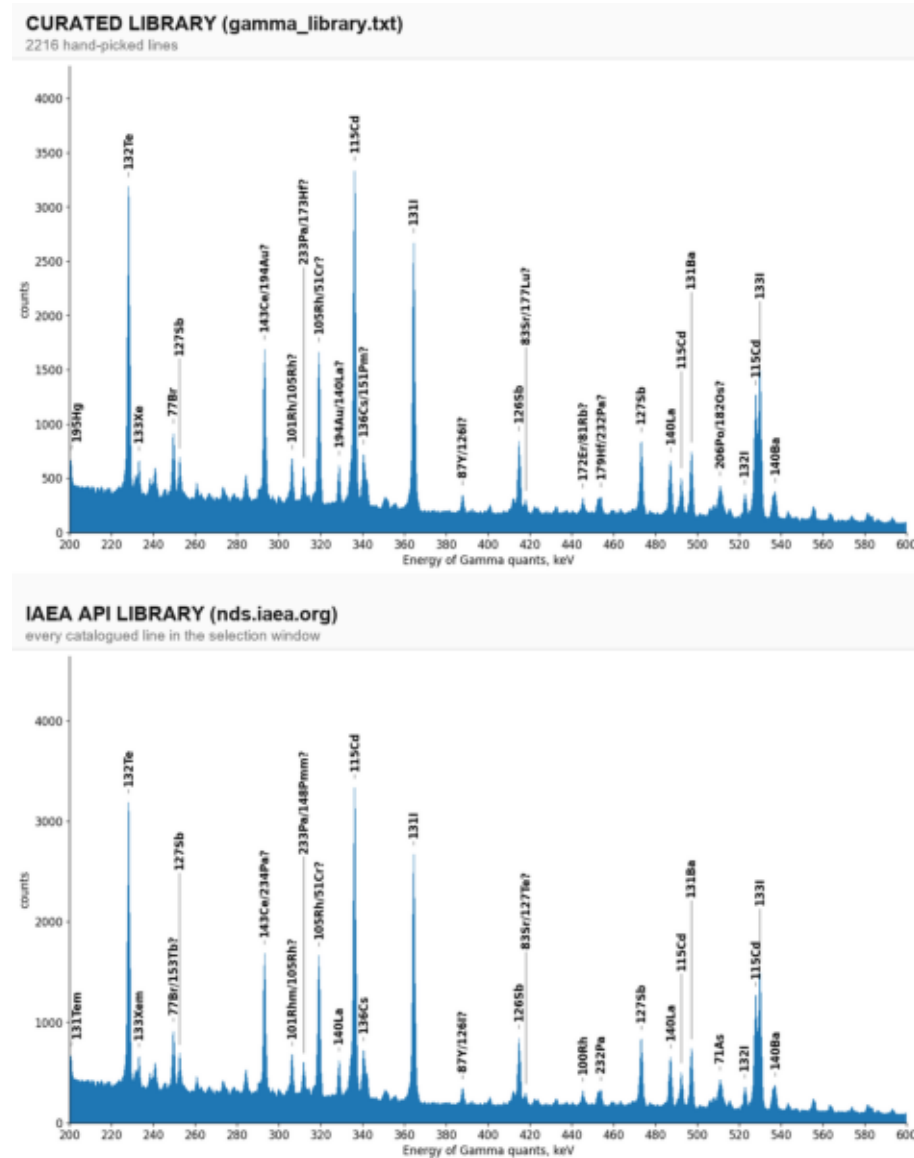
Spectrum figures: the bundled test_spectrum.Spe (1083 s acquisition, 5.04 days after a 119-minute proton irradiation), identified against the 50 538-line library assembled from per-nuclide decay radiation data. Chart figures: the isotope ^{81}Ga from the bulk ground-states table (3386 nuclides, one request).

Summary spectrum with IAEA API identifications, 200-600 keV



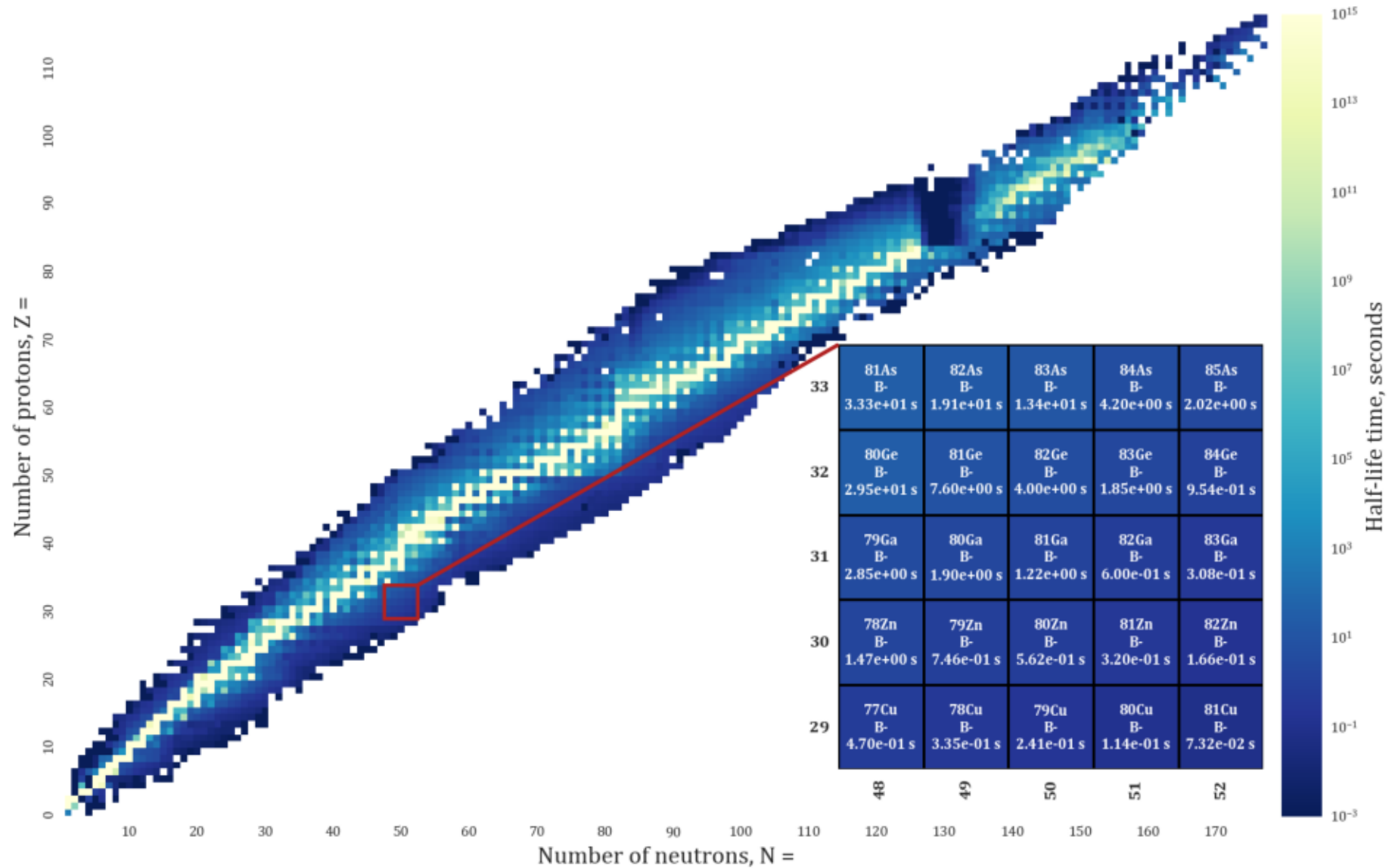
The same 59 detected peaks as the file-based analyzer -- every fit and activity number is identical; only the identification source differs. Metastable parent states are named explicitly (131Te, 101Rh) because the API reports the emitting state. The library passes through a post-pull curation stage: equilibrium half-life propagation through the feeding graph, and pinning of ambient background lines.

Curated library vs IAEA API library



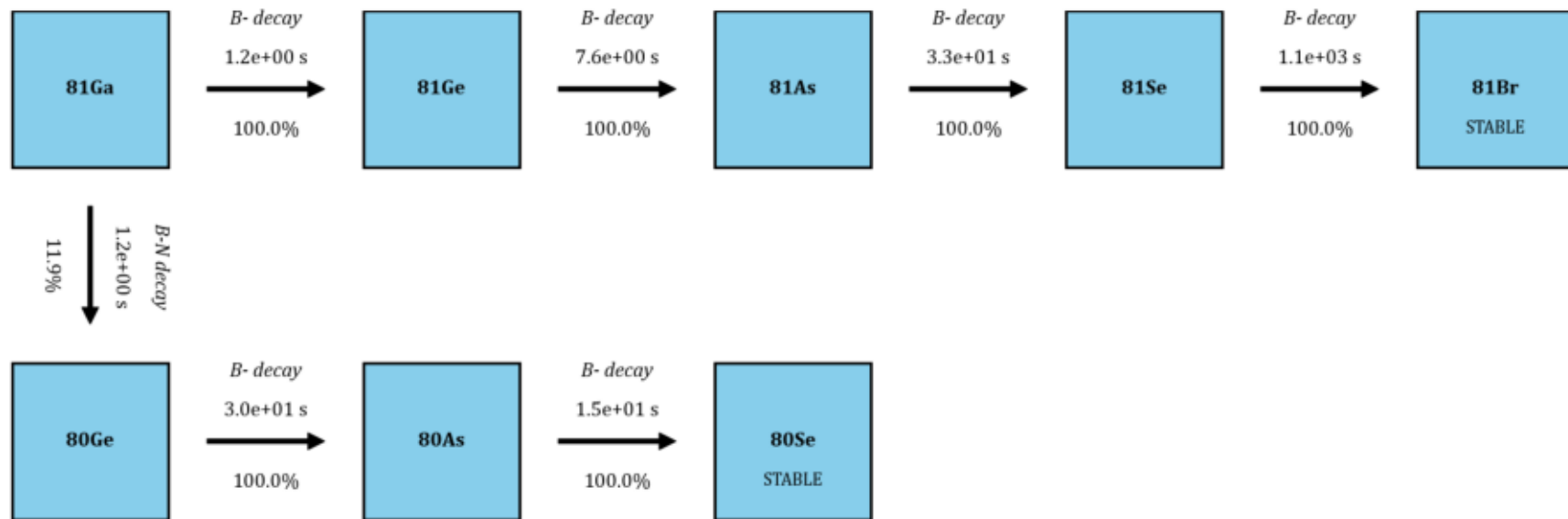
Identifications agree for 48 of 59 peaks. The curated library encodes transient equilibrium by listing generator-fed daughters under the rate-limiting parent half-life; the API library reproduces this with a computed post-pull stage that propagates feeder half-lives through the decay graph (99mTc 6 h -> 66 h via 99Mo, 140La -> 12.75 d via 140Ba) pins known ambient background lines such as 40K at 1461 keV, and keeps only parents reachable from a proton-irradiated 232Th target. Most remaining differences are fission products the complete line list identifies where the curated set had no entry, plus the actinide K X-rays and the 511 keV annihilation line.

Chart of nuclides from the ground-states table



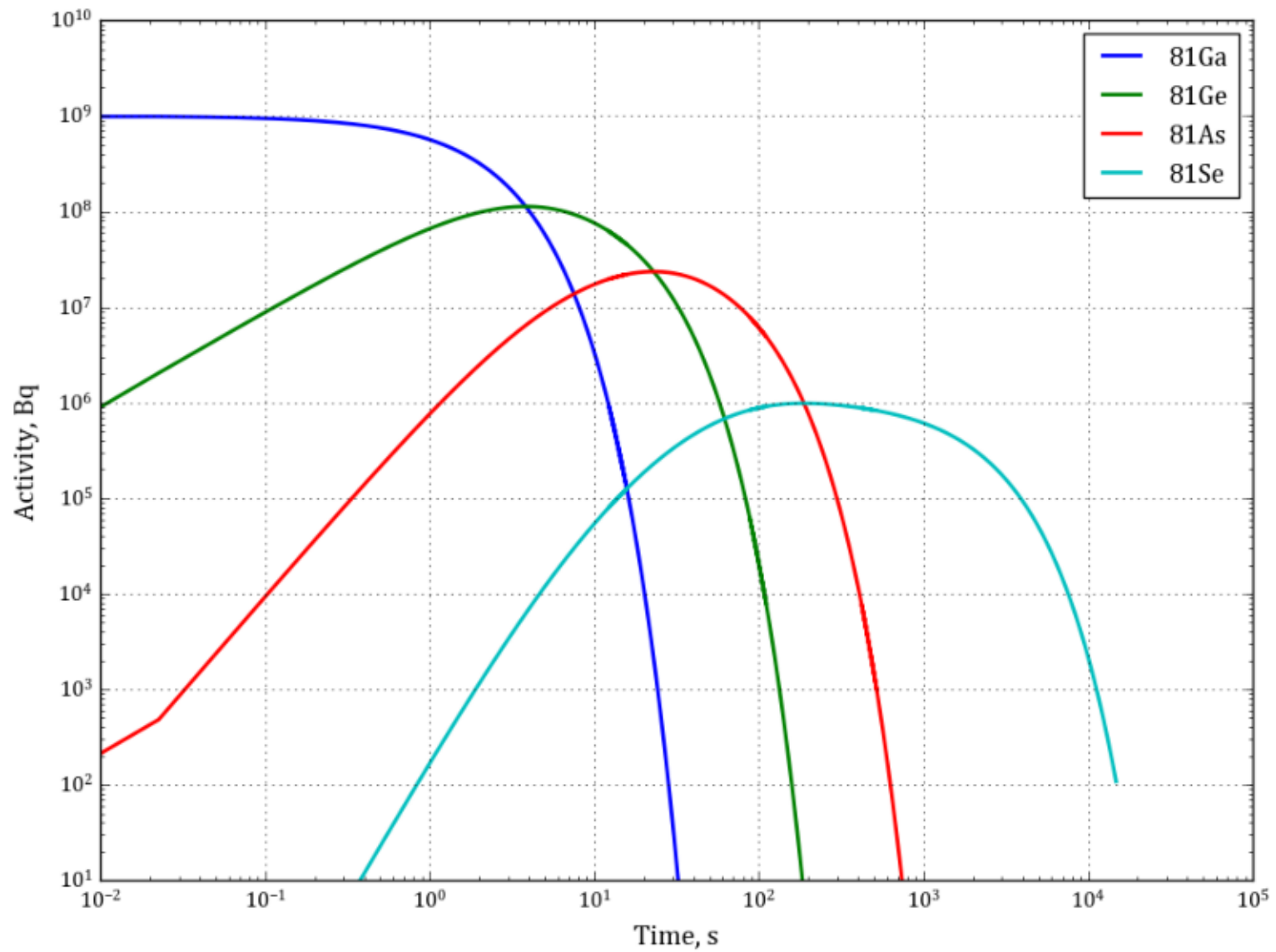
Half-life heatmap over all 3386 IAEA nuclides with the inset on 81Ga. Rendered by the same figure code as the database version; against the local data, half-lives agree within 5% for 90% of the 2820 common unstable nuclides, and several placeholder values from the 2013 NuDat export are replaced by measured ones.

Decay chain of 81Ga from API data



Identical member for member to the database version: the main chain $^{81}\text{Ga} > ^{81}\text{Ge} > ^{81}\text{As} > ^{81}\text{Se}$ to stable ^{81}Br , and the 11.9% beta-delayed-neutron branch to $^{80}\text{Ge} > ^{80}\text{As} > ^{80}\text{Se}$.

Activity of the chain members



Bateman activity from 1 GBq of ^{81}Ga , computed by matrix exponential of the decay matrix built from API branching ratios and half-lives. Indistinguishable from the database version.